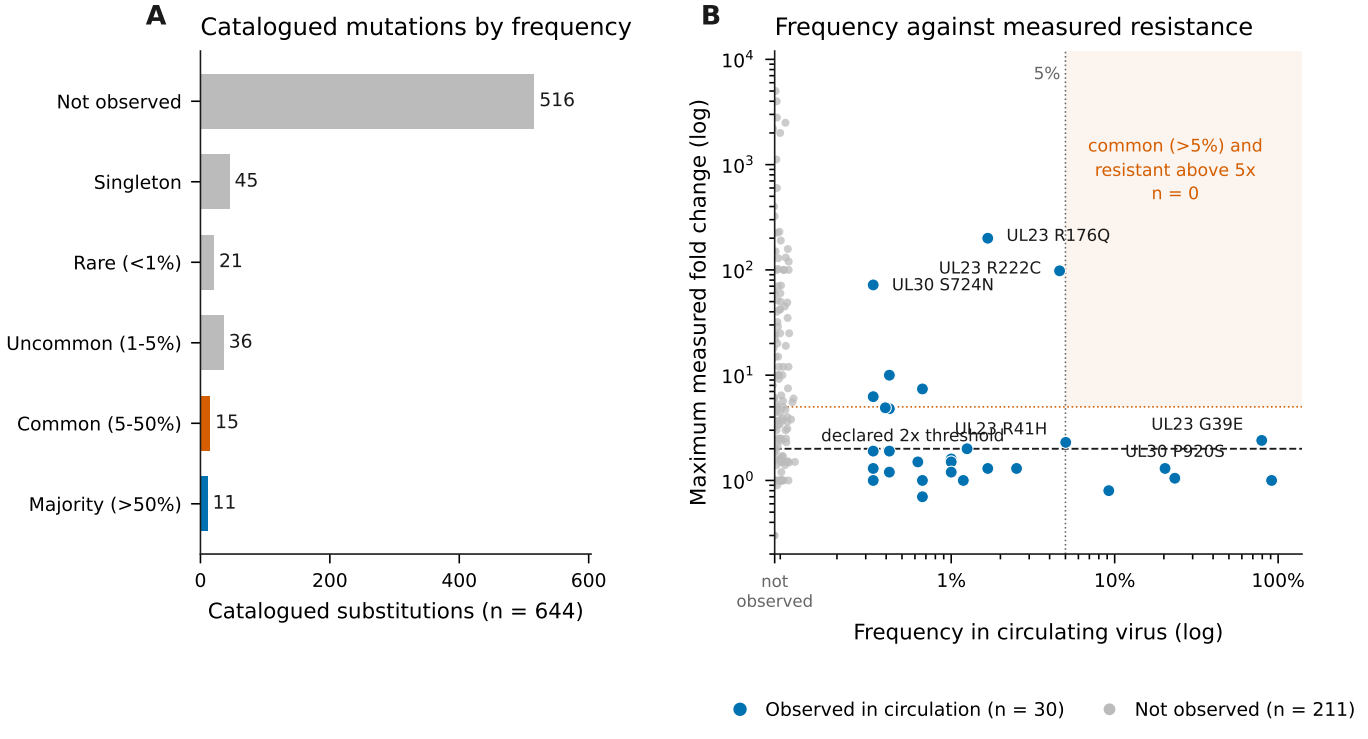
A
Catalogued mutations by frequency
Not observed
516
Singleton
45
Rare (<1%)
21
Uncommon (1-5%)
36
Common (5-50%)
15
Majority (>50%)
11
0
200
400
600
Catalogued substitutions (n = 644)
B
Frequency against measured resistance
Maximum measured fold change (log)
$10^4$
$10^3$
$10^2$
$10^1$
$10^0$
5%
common (>5%) and
resistant above 5x
n = 0
UL23 R176Q
UL23 R222C
UL30 S724N
declared 2x threshold
UL23 R41H
UL23 G39E
UL30 P920S
not
observed
1%
10%
100%
Frequency in circulating virus (log)
Observed in circulation (n = 30)
Not observed (n = 211)